



Enhancing Decision Tree Learning via Conditional Cumulative Residual Entropy: A Framework for Continuous Variables

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Abstract:

Conventional decision tree algorithms struggle with continuous variables, often causing information loss through discretization. This paper proposes a novel framework using Conditional Cumulative Residual Entropy (CCRE) to directly handle continuous inputs and outputs. Unlike traditional entropy measures, CCRE provides a more nuanced uncertainty measure for continuous distributions. The theoretical framework for the splitting criterion is derived, and its effectiveness is shown across regression and classification tasks. Empirical evaluations indicate that this approach eliminates discretization biases and yields superior predictive performance compared to baseline methods. The results highlight CCRE's potential as a powerful tool for tree-based learning, contributing significantly to data mining and machine learning.

Keywords: Conditional Cumulative Residual Entropy, Decision tree, Machin learning

Mathematics Subject Classification (2010): 99X99, 99X99, 99X99.

1 Introduction

In the field of machine learning and data mining, datasets generally consist of discrete input and output data, continuous input and output data, or combinations thereof. This diversity of data

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types presents various challenges in the design and application of learning algorithms. Decision trees are an effective method for classification and regression tasks, applicable to both discrete and continuous data. However, many decision tree algorithms are originally designed for discrete data, facing limitations when dealing with continuous variables. These limitations often necessitate discretization, which can lead to significant loss of valuable information. Abolhosseini et al. (2024) introduced an algorithm that uses cumulative residual entropy to improve decision tree methods, primarily focusing on datasets with discrete inputs and outputs. In contrast, this article specifically addresses datasets with continuous inputs and outputs, proposing a novel decision tree algorithm based on conditional cumulative residual entropy. This algorithm effectively manages continuous data without the need for discretization. The proposed algorithm demonstrates competitive performance against conventional and advanced methods in both regression and classification tasks, offering a promising alternative for handling complex and continuous data.

2 CCRDT algorithm

In the context of a scientific study, consider the target variable along with the features and the associated training dataset, which consists of n observations

$$\begin{bmatrix} y_1 & x_{11} & x_{21} & \cdots & x_{k1} \\ y_2 & x_{12} & x_{22} & \cdots & x_{k2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_n & x_{1n} & x_{2n} & \cdots & x_{kn} \end{bmatrix}$$

Furthermore, let us consider that X and Y are continuous random variables. In the newly proposed decision tree algorithm, referred to as CCRDT, the information gain criterion is determined using the CCRE in the following manner:

Information gain for feature X_i is defined as:

$$InformationGain(X_i) = CRE(Y) - CRE(Y|X_i); \quad i = 1, \dots, k. \quad (2.1)$$

where $CRE(Y) = -\sum_{i=1}^n P(Y \geq y_i) \log P(Y \geq y_i)(y_i - y_{i-1})$ is the cumulative residual entropy of the target variable in the training data set, and $CRE(Y|X_i)$ is the conditional cumulative residual entropy, calculated as follows for every X_i :

$$CRE(Y|X_i) = -\sum_{t=1}^n \sum_{j=1}^n \frac{\#(Y > y_t, X_i = x_{ij})}{n} \log \frac{\#(Y > y_t, X_i = x_{ij})}{\#(X_i = x_{ij})} (y_t - y_{t-1}) \quad (2.2)$$

The process of constructing a decision tree involves several key steps:

1. Calculate the conditional risk estimate (CRE) of the target variable, denoted as $\widehat{\varepsilon(Y)}$, using equation (1).
2. Determine the conditional CRE for each qualitative feature random variable through the estimator outlined in equation (2.2).
3. Assess the information gain associated with each feature by applying equation (2.1).
4. The feature that yields the highest information gain will be chosen as the splitting criterion at the tree's root. Subsequently, the training dataset will be divided based on the number of levels at the root, and the process from Steps 1 to 4 will be repeated until the final decision tree is fully developed.

3 Comparisons and Application to real data sets

Evaluating decision tree performance involves more than just predictive accuracy; computational efficiency and interpretability are also key. In this study, we examined the proposed new model (CCRDT) across a diverse set of datasets to ensure a comprehensive evaluation. All comparisons and calculations have been carried out systematically. In this article, as representative examples, the plots (figures) corresponding to two datasets are presented to visually illustrate the model's behavior. For the remaining datasets, a summary of the results is reported in Tables 1 and 2. Specifically, this study compares ID3 and CCRDT using accuracy, confusion matrix, F-score, and ROC curve. Random Forest, CCRDT, and CART are compared using MSE, RMSE, R-squared, and training time. Eight diverse datasets (e.g., Abalone, Concrete, Wine Quality, California Housing) were preprocessed and evaluated using cross-validation to ensure robustness and reduce overfitting. Finally, the proposed CCRDT algorithm is developed and applied to each dataset following outlined steps.

4 Conclusion

In contrast to conventional decision tree algorithms, the Conditional Cumulative Residual Decision Tree (CCRDT) eliminates the need for discretization by utilizing conditional cumulative residual entropy. This unique characteristic facilitates the development of decision trees that utilize the entirety of the information present in the dataset. As a result, the outcomes derived from CCRDT

Table 1: Comparative analysis of evaluation criteria for CCRDT and ID3

Dataset name		Evaluation criteria			
		Accuracy	Precision	Recall	F1-score
Abalone	CCRDT	0.76	0.80	0.68	0.74
Abalone	ID3	0.73	0.73	0.74	0.74
Concrete Compressive Strength	CCRDT	0.82	0.78	0.85	0.81
Concrete Compressive Strength	ID3	0.81	0.80	0.86	0.82
Wine Quality	CCRDT	0.53	0.53	0.99	0.70
Wine Quality	ID3	0.72	0.70	0.73	0.71
California Housing	CCRDT	0.79	0.78	0.89	0.83
California Housing	ID3	0.78	0.77	0.88	0.82
Energy Efficiency	CCRDT	0.99	0.97	1.00	0.99
Energy Efficiency	ID3	0.98	0.50	1.00	0.68
Airfoil Self-Noise	CCRDT	0.76	0.72	0.89	0.80
Airfoil Self-Noise	ID3	0.61	0.57	0.65	0.61
Combined Cycle Power Plant	CCRDT	0.93	0.98	0.90	0.94
Combined Cycle Power Plant	ID3	0.93	0.98	0.90	0.94
Life Expectancy (WHO) Fixed	CCRDT	0.91	0.94	0.90	0.92
Life Expectancy (WHO) Fixed	ID3	0.80	0.75	0.91	0.64

Table 2: Comparison between CCRDT, Random Forest and CART

Dataset name	Type of Tree	MSE	RMSE	R-squared	Training time (Second)
Abalone	CCRDT	1.07	1.03	0.30	2.19
	Random Forest	4.60	2.10	0.56	9.49
	CART	0.13	0.35	0.11	0.08
Concrete Compressive Strength	CCRDT	0.003	0.062	0.97	2.09
	Random Forest	0.36	0.56	0.88	0.94
	CART	0.11	0.34	0.51	0.02
Wine Quality	CCRDT	0.01	0.11	0.44	0.63
	Random Forest	0.34	0.55	0.52	2.02
	CART	0.01	0.12	0.34	0.03
California Housing	CCRDT	0.14	0.37	0.54	1.10
	Random Forest	1.20	2.06	0.70	0.36
	CART	0.20	0.39	0.56	0.18

more accurately reflect the genuine underlying relationships within the data, unhindered by the limitations imposed by discretization.

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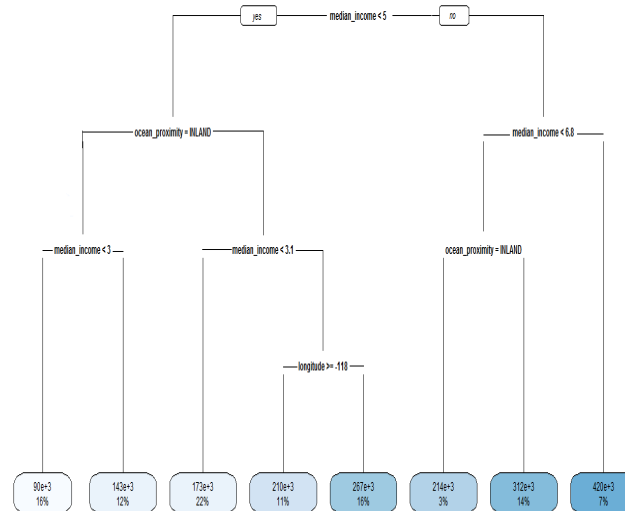


Figure 1: The ID3 decision tree. California Housing (Pace & Barry, 1997).

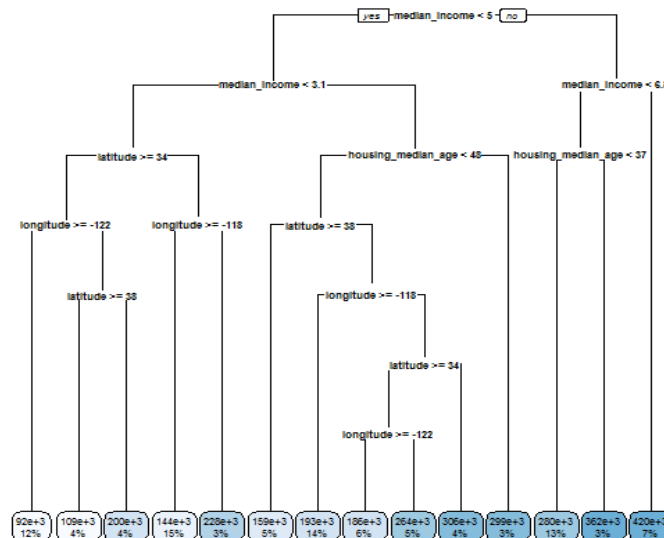


Figure 2: The CCRDT decision tree. California Housing (Pace & Barry, 1997).

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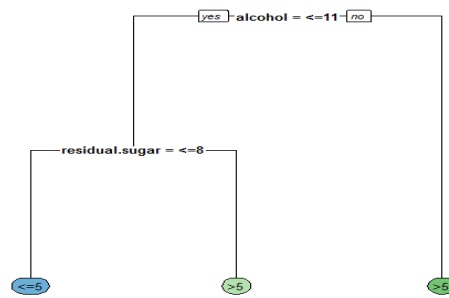


Figure 3: The ID3 decision tree. Wine Quality (Cortez et al., 2009).

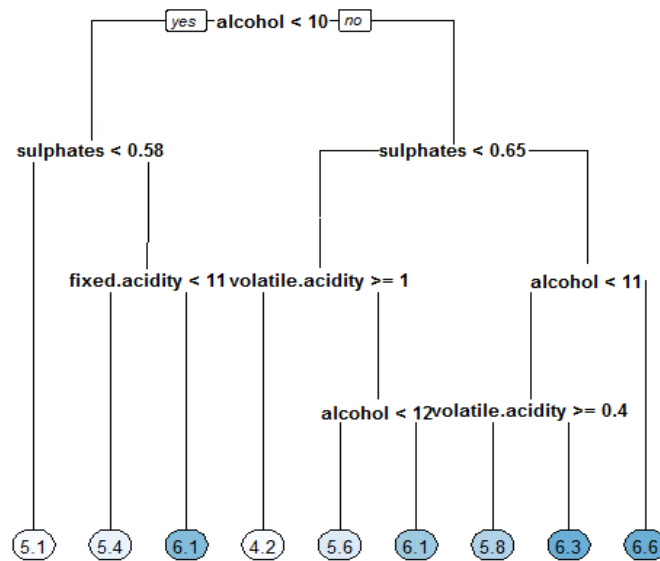


Figure 4: The CCRDT decision tree. Wine Quality (Cortez et al., 2009).

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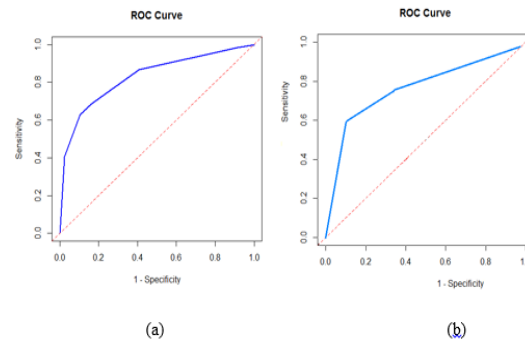


Figure 5: ROC curve of CCRDT (a) and ID3 (b) (California Housing).

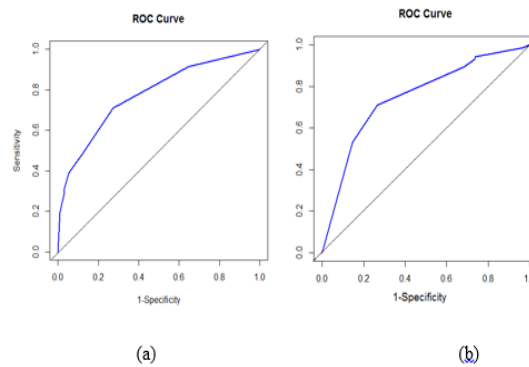


Figure 6: ROC curve of CCRDT (a) and ID3 (b) (Wine Quality).

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